

203. Remove Linked List Elements

PLATFORM

Platform: LeetCode

DIFFICULTY

EASY

DATE

Solved: June 22, 2026

Problem Summary

Given the head of a linked list and an integer val, remove all nodes whose value equals val and return the new head.

Approach

Use a dummy node pointing to head to avoid special-casing removal of the head itself. Walk through the list with curr, and whenever curr.val == val, unlink it via prev.next = curr.next; otherwise advance prev. curr always moves forward.

Algo

1. Create dummy node, dummy.next = head.
2. Set prev = dummy, curr = head.
3. While curr != null: if curr.val == val → prev.next = curr.next; else → prev = curr.
4. Move curr = curr.next.
5. Return dummy.next.

Key Insights

- Dummy node eliminates special-casing when head itself needs removal.
- prev only advances on non-matching nodes — handles consecutive duplicates correctly.
- curr always moves forward regardless of match/no-match.

Observations

- Same unlink pattern as LeetCode 83 (delete duplicates) — prev.next = curr.next skips a node without moving prev.
- Dummy node makes head removal as uniform as any other node removal.
- temp.next (not head) is returned, since head itself may have been removed.

Section

Time Complexity : $O(n)$ — single pass.

Space Complexity : $O(1)$ — only dummy, prev, curr pointers used.

The screenshot displays the LeetCode problem page for "203. Remove Linked List Elements". The problem description states: "Given the head of a linked list and an integer val, remove all the nodes of the linked list that have Node.val == val, and return the new head." Examples provided include: Example 1: Input: head = [1,2,6,3,4,5,6], val = 6; Output: [1,2,3,4,5]. Example 2: Input: head = [], val = 1; Output: []. Example 3: Input: head = [7,7,7,7], val = 7; Output: []. Constraints: The number of nodes in the list is in the range [0, 10^4]; 1 <= Node.val <= 50; 0 <= val <= 50. The solution is implemented in Java, using a dummy node and two pointers (prev and curr) to traverse the list and remove nodes with the specified value. The code is shown in the "Code" tab, and the "Testcase" tab shows the input and output for the first example.

203. Remove Linked List Elements Solved

Given the head of a linked list and an integer val, remove all the nodes of the linked list that have Node.val == val, and return the new head.

Example 1:
Input: head = [1,2,6,3,4,5,6], val = 6
Output: [1,2,3,4,5]

Example 2:
Input: head = [], val = 1
Output: []

Example 3:
Input: head = [7,7,7,7], val = 7
Output: []

Constraints:

- The number of nodes in the list is in the range [0, 10⁴].
- 1 <= Node.val <= 50
- 0 <= val <= 50

Seen this question in a real interview before? 1/8

Accepted 66 / 66 testcases passed
Sahil Khan submitted at Jun 22, 2026 17:49

Runtime
1 ms Beats 96.35%

Memory
47.38 MB Beats 39.41%

Code Java

```
1 /**  
2  * Definition for singly-linked list.  
3  * public class ListNode {  
4  *     int val;  
5  *     ListNode next;  
6  *     ListNode() {}  
7  *     ListNode(int val) { this.val = val; }  
8  *     ListNode(int val, ListNode next) { this.val = val;  
9  * }
```

Testcase Test Result

Accepted Runtime: 0 ms

Case 1 Case 2 Case 3

Input

head = [1,2,6,3,4,5,6]

val = 6